

Arduino Hand Wash Timer

Arlyn Benozza and Kenneth Manuntag



Kalp Patel
Varun Narravula

Drum Hero

(The better Guitar Hero)



Disclaimer: Don't report us for copying. We thought of it first, but they just made it before us

About Us:

Varun - Freshman here at Ohlone

Kalp - Sophomore transferring in Fall

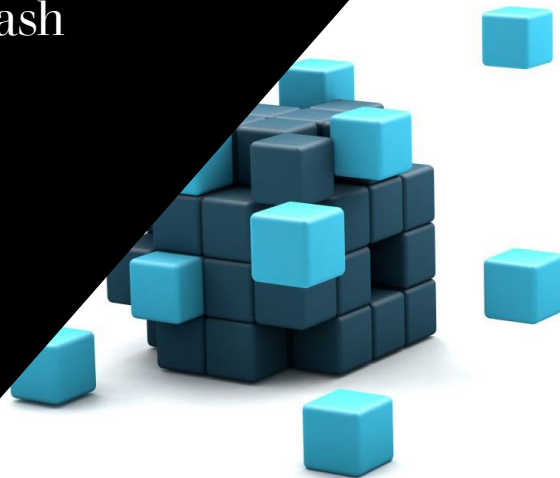
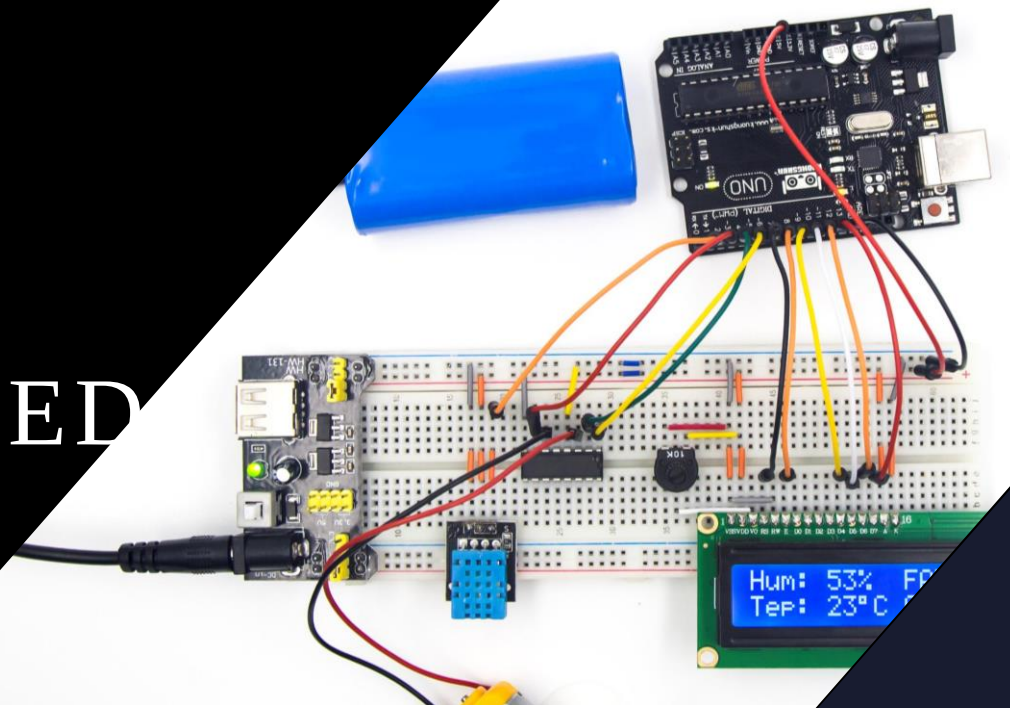
Ask us if you have any questions.

HUMIDITY AND TEMPERATURE CONTROLLED FAN

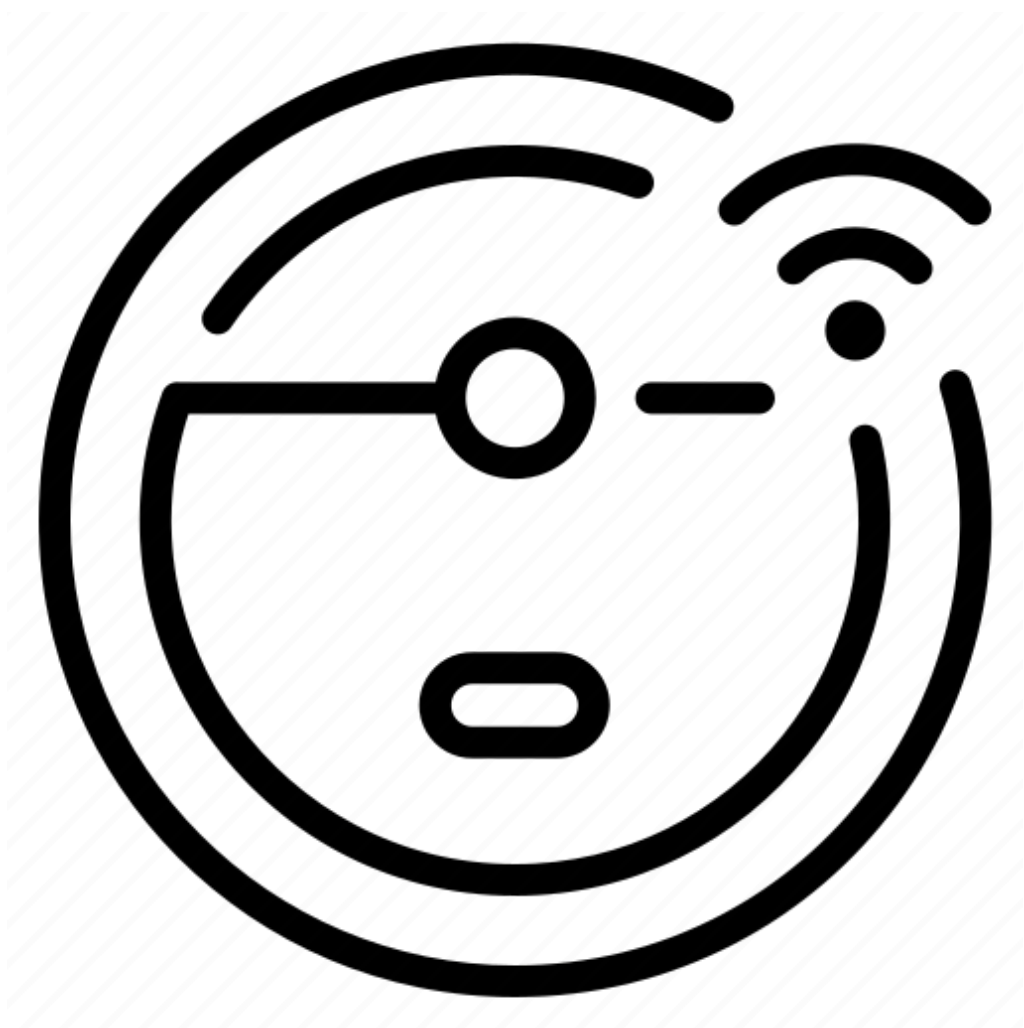
A&P TEAM

Abdulrahman Abuammash

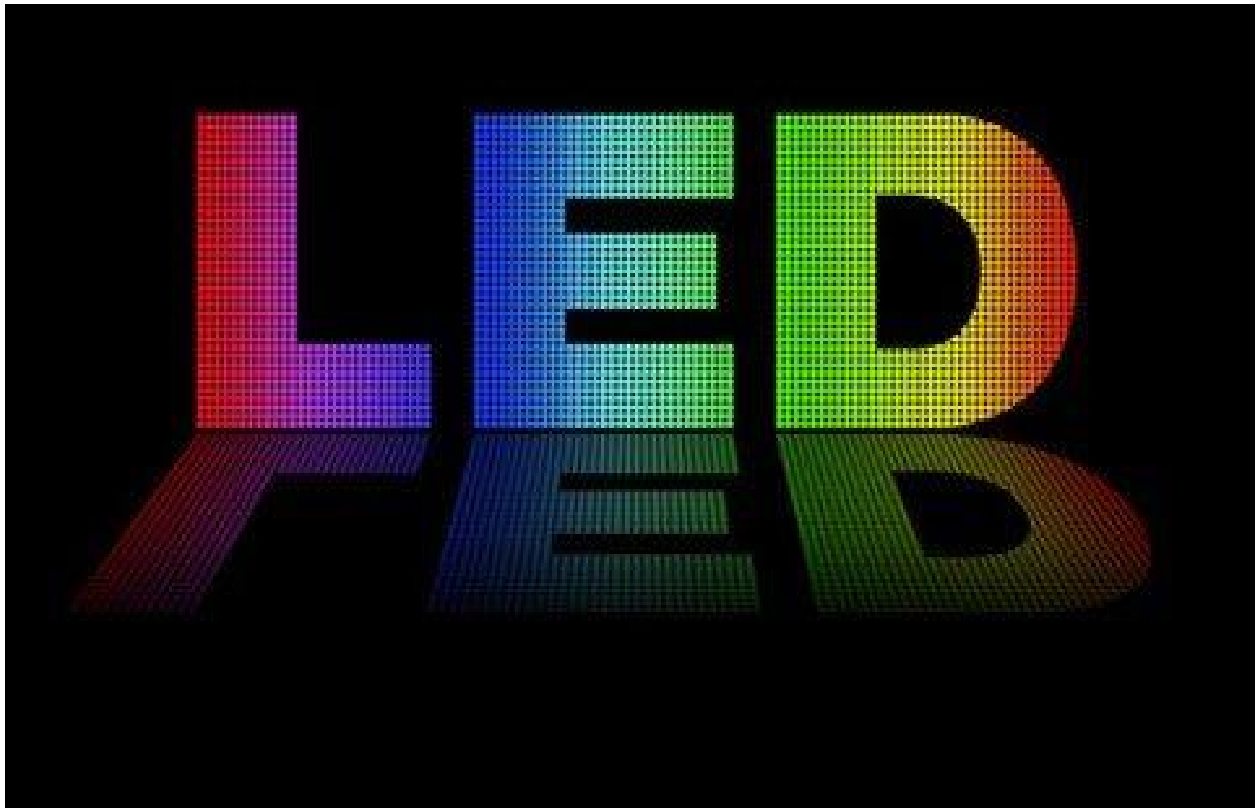
Pramodya Rajapakse



iRobot



Ying Zhou



LED Light Show

Team Shining Lights

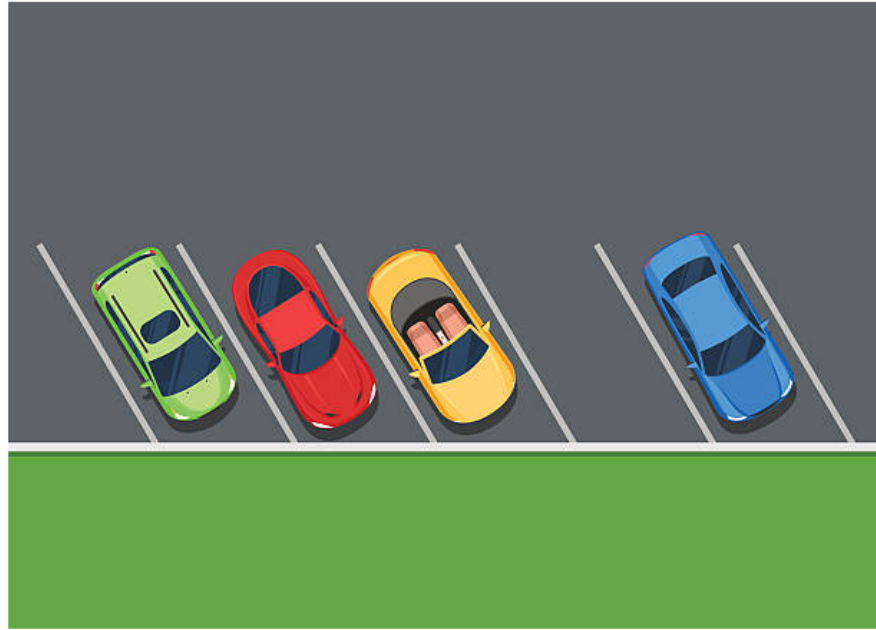
05.13.2022

Jotsaroop & Abdarrahan

Between 10:00am - 10:45am

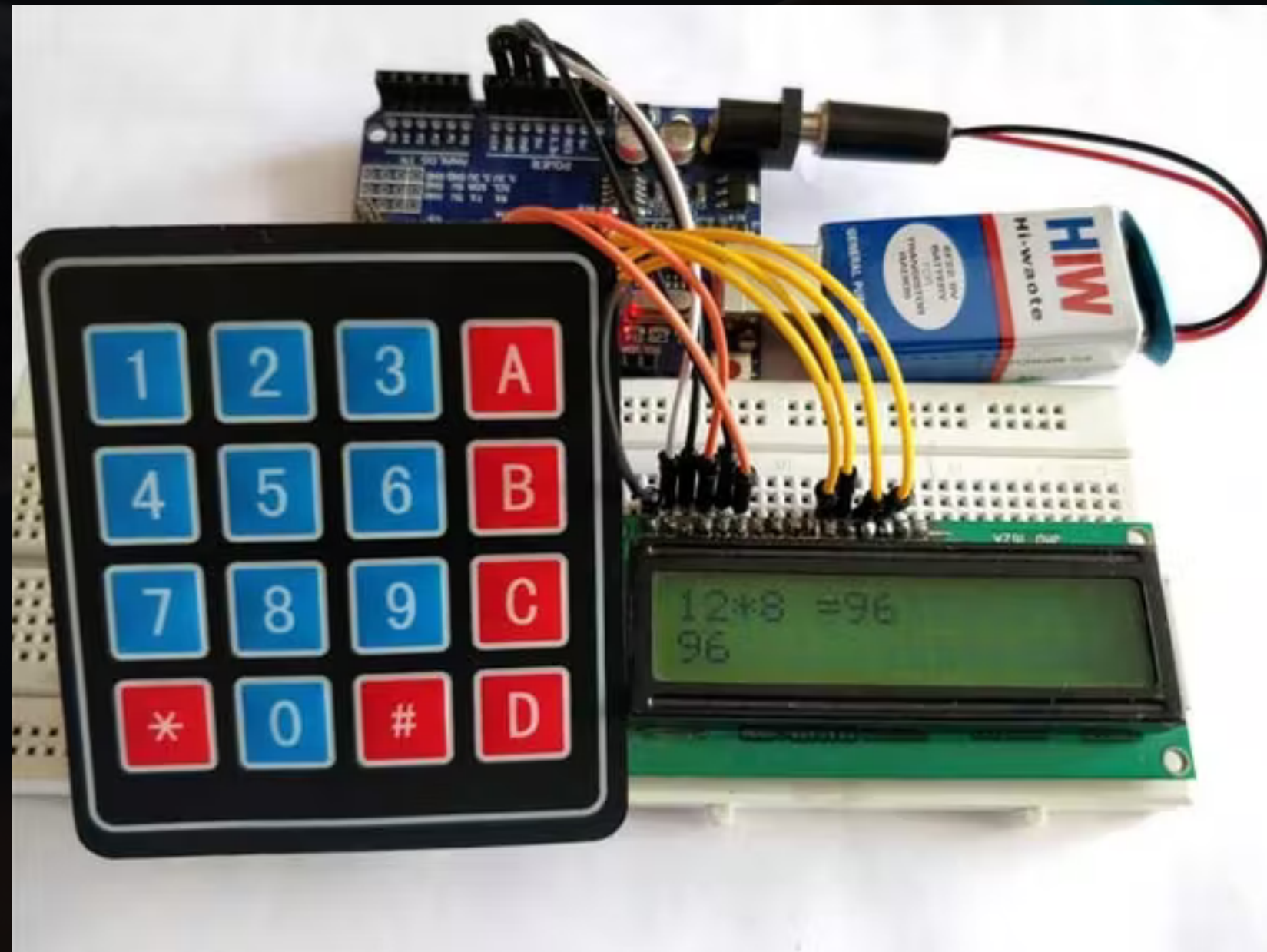
Fremont Campus Building 1, Floor 2

Parking Lot Arduino



Anthony La and Hamza Ahmed

SIMPLE CALCULATOR

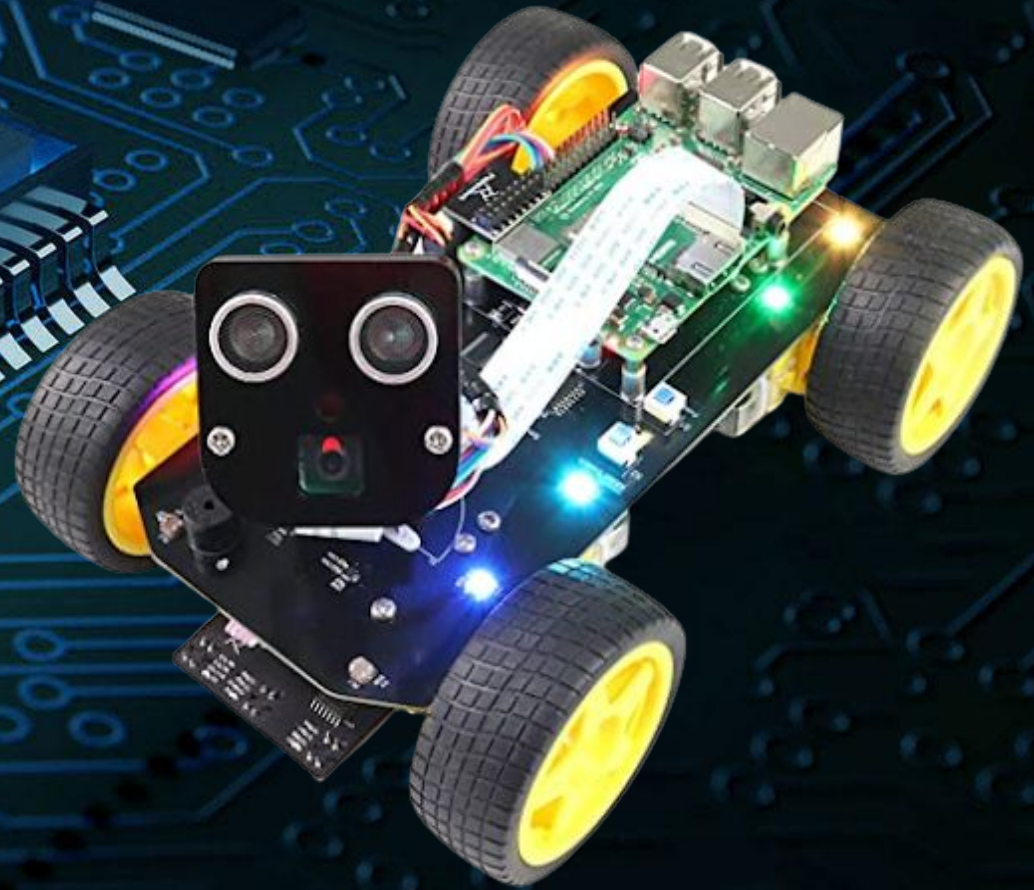


B A O T R U O N G
N G O C H U Y N H

Smart Car Kit with Raspberry Pi Project

Team Speedy

By Su Lei Aye
David Yu



Smart Home Modifier



Team Name: TLM

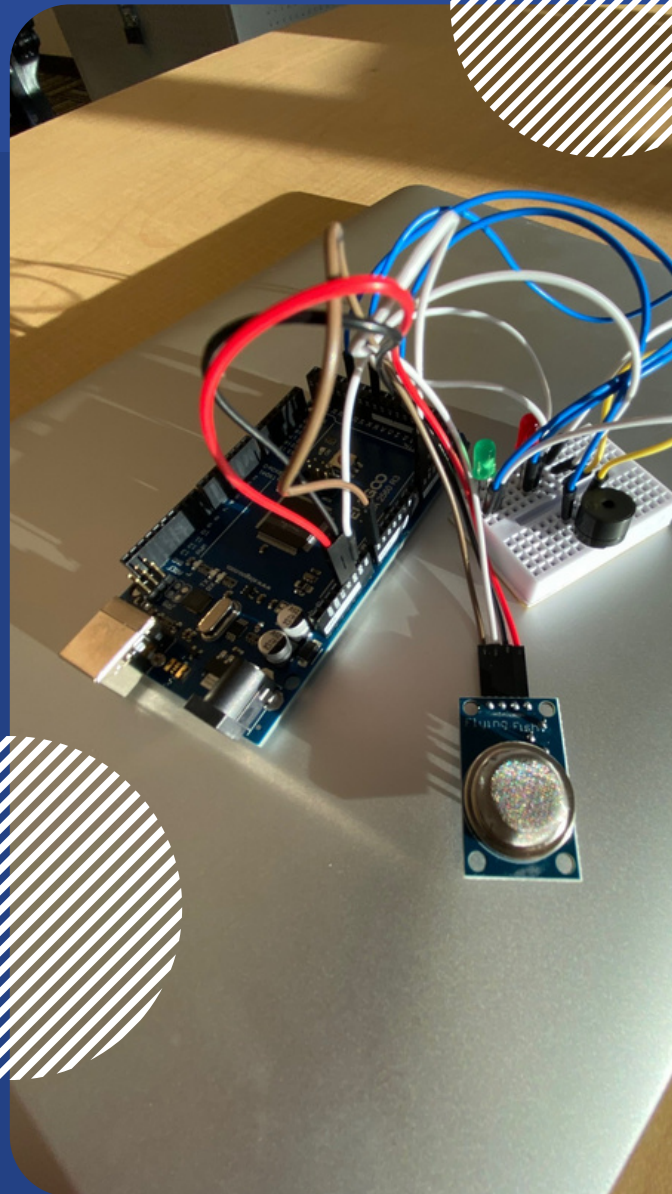
*Christopher
&
Evan*

SMOKE DETECTOR

JASSAN, NHAN

TEAM J&N

USING ARDUINO

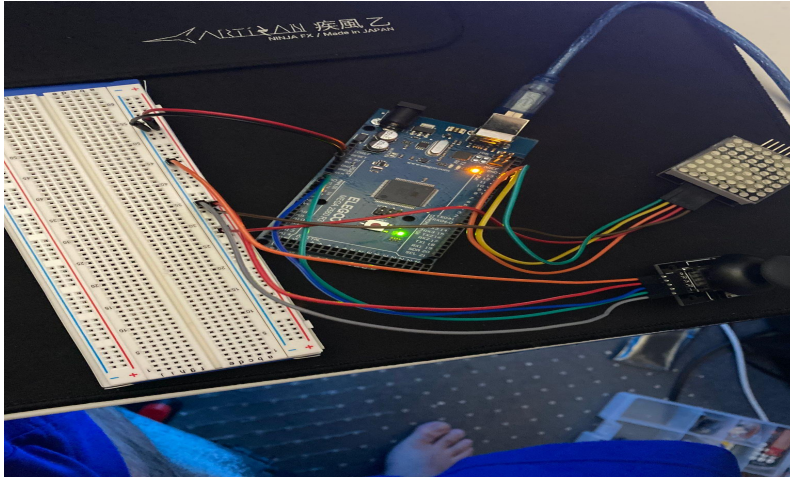




Snake Game

Brandon Villiados
Nicholas Jozovich-McNiel

First ProtoType



- Arduino Uno(Mega 2560 Controller Board),
- Analog joystick (generic),
- Breadboard (generic)
- LED Dot Matrix Display
- Female/Female Jumper Wires
- Male/Female Jumper Wires



Source Code(Struct of the snake)

```
#include <LedControl.h>

//Define The Snake as a Struct
typedef struct Snake Snake;
struct Snake{
    int head[2];    // the (row, column) of the snake head
    int body[40][2]; //An array that contains the (row, column) coordinates
    int len;        //The length of the snake
    int dir[2];     //A direction to move the snake along
};
```

- This shows how the snake starts and constructs throughout the game
- Shows the direction of which way the snake is able to move



Source code cont(struct of the apple)

```
//Define The Apple as a Struct
typedef struct Apple Apple;
struct Apple{
    int rPos; //The row index of the apple
    int cPos; //The column index of the apple
};
```

- This shows how the apple starts on a random row and column on the matrix



Source code cont(LED Matrix)

```
//MAX72XX led Matrix
const int DIN =12;
const int CS =11;
const int CLK = 10;
LedControl lc = LedControl(DIN, CLK, CS,1);

const int varXPin = A3;//X Value from Joystick
const int varYPin = A4;//Y Value from Joystick

byte pic[8] = {0,0,0,0,0,0,0,0};//The 8 rows of the LED Matrix

Snake snake = {{1,5},{0,5}, {1,5}}, 2, {1,0}};//Initialize a snake object
Apple apple = {(int)random(0,8),(int)random(0,8)};//Initialize an apple object

//Variables To Handle The Game Time
float oldTime = 0;
float timer = 0;
float updateRate = 3;
```

- Chose A3 on the Arduino mega 2560 for the x value of the joystick
- Chose A4 on the Arduino mega 2560 for the y value of the joystick
- Snake snake is to get the actual snake on the LED matrix
- Apple apple is to have the apple show up randomly on the LED matrix



Source Code

```
void Update()
{
    reset();//Reset (Clear) the 8x8 LED matrix

    int newHead[2] = {snake.head[0]+snake.dir[0], snake.head[1]+snake.dir[1]};

    //Handle Borders
    if(newHead[0]==8){
        newHead[0]=0;
    }else if(newHead[0]==-1){
        newHead[0] = 7;
    }else if(newHead[1]==8){
        newHead[1]=0;
    }else if(newHead[1]==-1){
        newHead[1]=7;
    }

    //Check If The Snake hits itself
    for(j=0;j<snake.len;j++){
        if(snake.body[j][0] == newHead[0] && snake.body[j][1] == newHead[1]){
            //Pause the game for 1 sec then Reset it
            delay(1000);
            snake = {{1,5},{0,5}, {1,5}}, 2, {1,0}};//Reinitialize the snake object
            apple = {(int)random(0,8),(int)random(0,8)};//Reinitialize an apple object
            return;
        }
    }
}
```

- Check if snake hits itself
 - If true game will pause for “x” amount of time then reset to default settings



Source Code

```
//Check if The snake ate the apple
if(newHead[0] == apple.rPos && newHead[1] == apple.cPos){
    snake.len = snake.len+1;
    apple.rPos = (int)random(0,8);
    apple.cPos = (int)random(0,8);
}else{
    removeFirst();//Shifting the array to the left
}

snake.body[snake.len-1][0] = newHead[0];
snake.body[snake.len-1][1] = newHead[1];

snake.head[0] = newHead[0];
snake.head[1] = newHead[1];

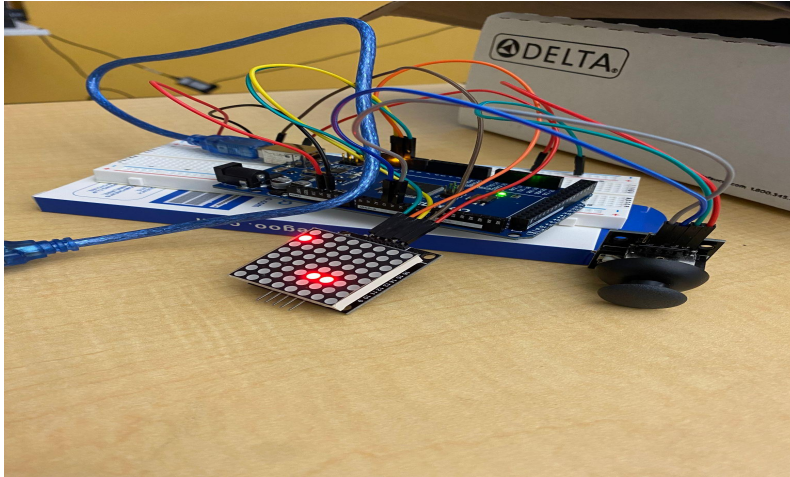
//Update the pic Array to Display(snake and apple)
for(j=0;j<snake.len;j++){
    pic[snake.body[j][0]] |= 128 >> snake.body[j][1];
}
pic[apple.rPos] |= 128 >> apple.cPos;
}

void Render()
{
    for(i=0;i<8;i++){
        lc.SetRow(0,i,pic[i]);
    }
}

void removeFirst()
{
    for(j=1;j<snake.len;j++){
        snake.body[j-1][0] = snake.body[j][0];
        snake.body[j-1][1] = snake.body[j][1];
    }
}
```

- Check if snake eats the apple
 - If true the snake continues to grow
- Shows how the snake and apple are displayed on the LED matrix

Second Prototype



- Got the LED Matrix to work
 - Had issues with the LED control in the libraries previously
- Got the analog joystick to work
- Got the snake game to work correctly



Features and Quantifiers

Function to generate food for snake: domain of food row $-1 < x < 9$. Domain of food column $-1 < y < 9$. Where x and y are random and not generated where snake is generated.

Snake length $z = 3$ where $\text{snake}[\text{pos}(\text{head})] = x = 3$. $\text{Snake}[\text{pos}(\text{tail})] = x = 3-2$. Snake starts at the same position every time in the middle of the matrix where the snake food is randomized but cannot be generated on top of snake.